

Weekly Progress Report

Refactoring Detection Project - Week 7 progress update for the Monday research supervisor meeting

1. Overview

During Week 7, the team moved from branch integration into final UI cleanup and research-paper development. The main work was reviewing the fully integrated CLONE interface, fixing layout and labeling problems, adding scrollbars so the complete panel remains usable inside IntelliJ, and documenting the project in Overleaf. Eight of the eleven identified UI issues were resolved, leaving three final fixes for the next work period.

The team also kept the project website and shared files up to date. This week marks a transition from implementation and integration toward final verification, documentation, and preparation of the complete research report.

2. Weekly Progress Summary

Area	Progress Made
UI fixes	Eight of eleven identified issues in the integrated UI Panel were fixed.
UI improvements	Scrollbars and layout corrections were added so panel sections flow correctly without overlap or hidden content.
Branding and labels	Remaining AntiCopyPaster wording in the waiting state was changed to CLONE, and duplicate or unnecessary confidence/help text was removed.
Overleaf	The team learned the Overleaf workflow and began writing the research paper, including the abstract, introduction, problem statement, and new-features sections.
Website and files	The website remained current and all weekly materials were uploaded and organized through OneDrive.

3. UI Improvements and Fixes

The integrated panel was reviewed as a complete IntelliJ tool window instead of as separate feature branches. The review identified eleven issues related to labels, spacing, dropdown behavior, scrolling, button placement, and overall visibility. The following eight fixes were completed:

Fix	Completed Change
Branding update	Changed the waiting-state label from [AntiCopyPaster] to [CLONE] inside showWaitingState so the final interface uses consistent project branding.
Help button cleanup	Removed the extra question mark from the help button in the action panel.
Multiple-clone dialog alignment	Shifted text in CloneOccurrenceDialog so the message aligns correctly with its label on the left side of the dialog.
Confidence label cleanup	Removed Confidence: Pending from the Details & Provenance dropdown label because the dedicated AI Trust section already communicates validation status.
Panel scrolling	Wrapped the suggestion panel in a JScrollPane, adding scrollbars so users can reach all content without sections overlapping or becoming hidden.
Action-panel layout	Fixed createActionPanel layout problems that placed buttons beside the status label; the buttons now remain below the rest of the content.
IntelliJ-style dropdown icon	Replaced text arrows such as v and > in the Details & Provenance control with the IntelliJ dropdown-arrow icon.
Description spacing	Reduced spacing in the top description area to create more room for interactive features and make the panel cleaner.

4. Impact of the UI Changes

Improvement Area	Effect on the Final Panel
Consistency	The CLONE name now appears more consistently across the panel, reducing confusion between the old and final project identity.
Visibility	Scrollbars allow users to access the explanation, diff, trust details, status messages, and action buttons even when the IntelliJ tool window is narrow.
Clarity	Removing duplicate confidence text and the extra help symbol makes the interface easier to scan.
IntelliJ familiarity	The dropdown icon and layout behavior better match normal IntelliJ interface patterns.
Usability	Correct button placement and reduced spacing keep the most important actions available without covering other content.

5. Overleaf and Research Paper Progress

The team began preparing the final research paper in Overleaf. The initial writing explains the motivation for improving CLONE, the problems in earlier interface versions, the new UI features, and the role of human review in AI-assisted refactoring.

Paper Section	Week 7 Progress
Abstract	Introduced the CLONE plugin, summarized its main features, and stated the goal of the research project.

Paper Section	Week 7 Progress
Introduction	Explained why the CLONE plugin is important, outlined how the plugin works, summarized the UI improvements, and introduced the paper structure.
Problem Statement	Described the limitations of previous CLONE interface iterations and explained the problems addressed by the redesigned panel.
New Features	Started documenting each new feature with images and explanations of the changes introduced during the project.

6. Feature Subsections Added to the Paper

Feature	Documentation Added
Multiple Clone Dialog	Introduces the popup shown when multiple clones are detected and explains the changes made to the occurrence-selection interface.
Explanation Section	Explains the purpose of the explanation area, the information it should contain, and why clone type, locations, line numbers, and suggested method details matter.
Diff Preview	Describes the role of the before-and-after code view, its importance for reviewing refactorings, and the complications addressed by the redesigned presentation.
AI Trust Panel	Documents the five equally weighted validation stages and the verified, pending, and failed interface states.
Action Buttons	Describes the improved Edit action, the conditions that disable Apply, and the human-in-the-loop workflow for accepting or rejecting changes.

7. Remaining Work and Next Steps

- Complete the final three UI issues and use the newly available API tokens to view and verify the changes inside the full panel.
- Run the updated panel in IntelliJ and confirm that scrolling, spacing, buttons, dropdowns, and labels behave correctly together.
- Finish and expand the Overleaf paper, especially the workflow section and detailed descriptions of each feature.
- Add images of each individual UI section and an image of the complete integrated CLONE panel to the research paper.
- Continue updating the project website, OneDrive files, weekly reports, and supporting documents.

8. Conclusion

Overall, Week 7 focused on polishing the integrated CLONE interface and beginning the final research paper. The team resolved most of the identified UI problems, improved the tool window layout with scrolling and cleaner IntelliJ-style controls, and documented the main research contributions in Overleaf. The remaining work is to complete the last three interface fixes, verify the full panel, finish the paper, and add final screenshots that demonstrate the completed system.